

Solution Requirements

Date	27 june 2026
Team ID	
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Step 1: Functional Requirements (FR)

S.No.	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	User Input Collection	Allow farmers to input soil and environmental parameters such as N, P, K, temperature, humidity, pH level, rainfall, and crop type.	High
2	Smart Crop Recommendation	Analyze the input data and recommend the most suitable crop for maximum yield and better resource efficiency.	High
3	Crop Suitability Assessment	Evaluate whether current soil and climate conditions are suitable for a selected crop and provide compatibility insights.	High
4	Environmental Data Analysis	Analyze environmental factors and their impact on crop productivity and growth.	Medium

S.No.	Requirement Category	Requirement Description	Priority (High/Medium/Low)
5	Resource Optimization Recommendations	Provide intelligent recommendations for optimal use of resources (water, fertilizers, nutrients, etc.) to improve efficiency.	Medium
6	Data Visualization & Insights	Display insights and patterns through charts, graphs, and dashboards for better decision-making.	Medium
7	Historical Data Management	Store and manage historical data of soil, climate, and crop performance for future analysis.	Medium
8	Report Generation	Generate detailed reports and summaries for farmers, researchers, and policymakers.	Low
9	User Management	Manage user registration, authentication, and different roles (Farmer, Researcher, Admin, Policymaker).	Medium
10	Notifications & Alerts	Send alerts or notifications regarding abnormal conditions, weather changes, or recommendations.	Low

Step 2: Non-Functional Requirements (NFR)

S.No.	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	System should process input data and provide recommendations quickly.	Response time < 3 seconds
2	Scalability	System should handle increasing number of users and large datasets efficiently.	Supports up to 10,000 concurrent users
3	Security & Data Privacy	Ensure secure storage and transmission of user and agricultural data.	End-to-end encryption (AES-256)

S.No.	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
4	Reliability & Availability	System should have high uptime and minimal downtime.	99.9% uptime per month
5	Usability & Accessibility	System should be user-friendly and accessible on multiple devices.	Complies with WCAG 2.1 AA standards
6	Maintainability	System code should be modular and easy to maintain and update.	Code coverage > 80%
7	Compatibility	System should be compatible with major browsers and platforms.	Works on Chrome, Firefox, Edge, and mobile browsers
8	Other	System should maintain data integrity and provide audit logs for critical operations.	Audit logs available for all major operations

Notes:

- Priority: High (Must have), Medium (Should have), Low (Could have)
- All requirements must be measurable and testable.